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Oefeningenreeks 4A nr. 67.5

$$\int \frac{x}{x^4+2x^2+2}dx$$
$$= \int \frac{x}{x^4+2x^2+1+1}dx$$
$$= \int \frac{x}{(x^2+1)^2+1}dx$$

Substitutie: $t = x^2+1 \Rightarrow dt = 2xdx$

$$= \frac{1}{2}\int \frac{1}{t^2+1}dt$$
$$= \frac{1}{2}\,\mathrm{Bgtan}\,t + c$$
$$= \frac{1}{2}\,\mathrm{Bgtan}(x^2+1) + c$$

References